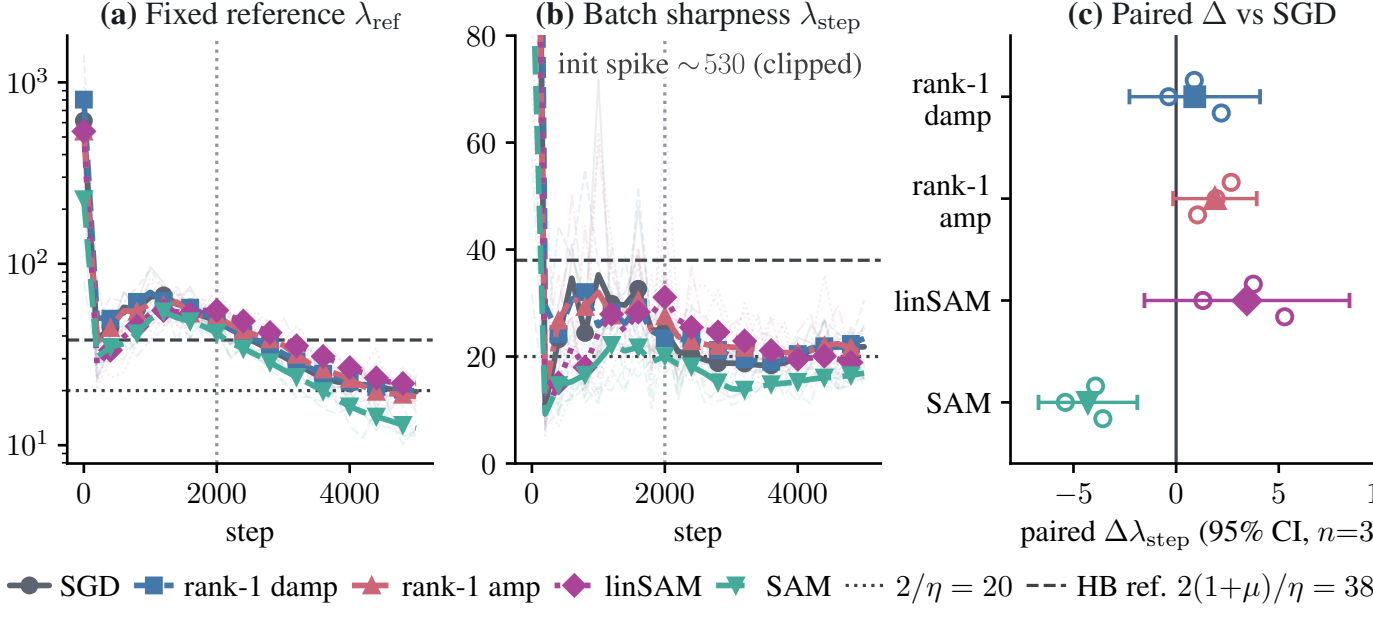

(a) Fixed reference λ_{ref}
10^3
10^2
10^1
0
2000
4000
step
(b) Batch sharpness λ_{step}
init spike $\sim$530 (clipped)
80
60
40
20
0
0
2000
4000
step
(c) Paired Δ vs SGD
rank-1 damp
rank-1 amp
linSAM
SAM
−5
0
5
10
paired $\Delta\lambda_{\text{step}}$ (95% CI, n=3)
SGD
rank-1 damp
rank-1 amp
linSAM
SAM
$2/\eta = 20$
HB ref. $2(1+\mu)/\eta = 38$